FBISE PHYSICS SSC-I

NEW PATTERN MODEL PAPER

COMPLETE SOLUTION

60 MCQs with reasoning | Section B all OR options | Section C all OR options

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Exam-oriented, step-by-step and student-friendly solution

Based on the Federal Board SSC-I Physics Model Question Paper
National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
50% MCQs + 50% Descriptive

Educational preparation material. This is not an official FBISE answer key.

How to Use This Solution

This booklet is designed for students preparing the FBISE SSC-I Physics model paper. It gives the correct option and reasoning for every MCQ, then provides exam-ready answers for every short and long question including all OR choices.

- For MCQs: read the reason, not only the answer letter.
- For numericals: write Given, Formula, Substitution, Calculation and Final Answer with unit.
- For theory questions: use clear definitions, key laws/formulae and a concluding statement.
- In the actual examination, attempt only the choices required by the paper. This booklet solves all OR options for preparation.

Paper Structure

Section	Marks	What this booklet provides
A	30	60 MCQs (0.5 mark each) with reasoning
B	18	6 short questions + all 6 OR alternatives
C	12	2 long questions + both OR alternatives

Section A - Quick Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	16	D	31	D	46	C
2	B	17	B	32	B	47	A
3	C	18	B	33	B	48	B
4	C	19	B	34	A	49	B
5	C	20	C	35	A	50	A
6	C	21	C	36	D	51	D
7	B	22	B	37	B	52	C
8	A	23	B	38	D	53	B
9	B	24	C	39	B	54	A
10	A	25	C	40	B	55	D
11	B	26	B	41	B	56	C
12	C	27	B	42	B	57	C
13	A	28	C	43	A	58	D
14	B	29	A	44	B	59	B
15	B	30	B	45	B	60	D

Important: Q41 uses the best matching option from the choices; the scenario states that the car speeds up but does not explicitly say acceleration is perfectly constant.

Section A - Detailed MCQ Solutions

Each question carries 0.5 mark in the model paper. The explanations below are intentionally fuller than what students need to write, so they can understand the concept behind each answer.

MCQ 1

Which of the following is a base quantity?

A. Velocity	B. Mass
C. Force	D. Momentum

Correct answer: B. Mass

Reasoning: Mass is one of the seven SI base quantities. Velocity (m/s), force (kg m/s^2) and momentum (kg m/s) are derived from base quantities.

Exam focus: Know the seven SI base quantities and distinguish them from derived quantities.

MCQ 2

The rate of change of displacement is called:

A. Speed	B. Velocity
C. Acceleration	D. Distance

Correct answer: B. Velocity

Reasoning: Velocity is the rate of change of displacement with time: $v = \Delta x/\Delta t$. Speed is the rate of change of distance, while acceleration is the rate of change of velocity.

Exam focus: Displacement/time gives velocity; distance/time gives speed.

MCQ 3

Which of the following forces is a non-contact force?

A. Friction	B. Tension
C. Magnetic force	D. Air resistance

Correct answer: C. Magnetic force

Reasoning: Magnetic force can act without physical contact. Friction, tension and air resistance require interaction with matter.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 4

The tendency of an object to resist a change in its state of rest or motion is called:

A. Force	B. Weight
C. Inertia	D. Momentum

Correct answer: C. Inertia

Reasoning: Inertia is the property of matter by which a body resists any change in its state of rest or uniform motion. Greater mass means greater inertia.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 5

Pascal's Law is applicable to:

A. Gases only	B. Solids only
C. Fluids	D. Vacuum

Correct answer: C. Fluids

Reasoning: Pascal's law states that pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions. Fluids include liquids and gases.

Exam focus: Pascal's law is the key principle behind hydraulic machines.

MCQ 6

Which of the following is a scalar quantity?

A. Force	B. Displacement
C. Energy	D. Velocity

Correct answer: C. Energy

Reasoning: A scalar has magnitude only. Energy is scalar, while force, displacement and velocity have both magnitude and direction.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 7

The SI unit of momentum is:

A. N m	B. kg m/s
C. kg m/s^2	D. J

Correct answer: B. kg m/s

Reasoning: Momentum p = mv, so its SI unit is kg x m/s = kg m/s.

Exam focus: Always derive units from the defining formula if unsure.

MCQ 8

Which statement best describes translational motion of a body?

A. All its parts move in one direction	B. Rotates about a fixed axis
C. Moves about a fixed point	D. Changes its shape

Correct answer: A. All its parts move in one direction

Reasoning: In translational motion, all parts of a body undergo displacement in the same direction. Rotation about a fixed axis is rotational motion.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 9

The value of 'g' at the surface of the Moon is approximately:

A. 9.8 m/s^2	B. 1.6 m/s^2
C. Infinite	D. 0 m/s^2

Correct answer: B. 1.6 m/s^2

Reasoning: The Moon's gravitational field strength is about one-sixth of Earth's: 9.8/6 ≈ 1.63 m/s^2, so 1.6 m/s^2 is the best answer.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 10

Two identical beakers contain the same amount of water. Beaker A is at 30 C while beaker B is at 60 C. Why?

A. Particles of beaker B have more internal energy	B. Particles of beaker A have more density
C. Particles of beaker A have more internal energy	D. Particles of beaker B have less density

Correct answer: A. Particles of beaker B have more internal energy

Reasoning: For the same amount of water, the higher-temperature sample has greater microscopic kinetic energy and therefore greater internal energy.

Exam focus: For the same substance and amount, higher temperature generally means greater internal energy.

Energy, Pressure and Mechanics

MCQ 11

A book is lifted from the floor to a shelf. Which factor is responsible for its gain in gravitational potential energy?

A. Energy lost by the book as heat	B. Work done against gravity to raise the book
C. Increase in the speed of the book	D. Increase in the weight of the book

Correct answer: B. Work done against gravity to raise the book

Reasoning: When the book is raised, work is done against gravitational force. That work appears as an increase in gravitational potential energy: $\Delta PE = mgh$.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 12

Which one of the following is a renewable energy source?

A. Coal	B. Natural gas
C. Solar energy	D. Nuclear energy

Correct answer: C. Solar energy

Reasoning: Solar energy is naturally replenished. Coal and natural gas are fossil fuels and are non-renewable on human timescales.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 13

Students carry a liquid barometer up a mountain and observe that the mercury level decreases. What is the reason?

A. Lower air pressure	B. Stronger gravity
C. Lower temperature	D. Higher air pressure

Correct answer: A. Lower air pressure

Reasoning: Atmospheric pressure decreases with altitude because there is less air above. A lower atmospheric pressure supports a shorter mercury column.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 14

The center of gravity of a uniform rod lies at:

A. One end	B. Its center
C. One-quarter from an end	D. Outside the rod

Correct answer: B. Its center

Reasoning: For a uniform rod, mass is evenly distributed, so the center of gravity lies at its geometrical center.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 15

What is the purpose of using ball bearings in a machine?

A. They increase friction	B. They reduce friction
C. They increase mass	D. They reduce air resistance

Correct answer: B. They reduce friction

Reasoning: Ball bearings replace sliding friction with smaller rolling friction, reducing wear, heating and energy loss.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 16

A file size is 2 gigabytes (GB). If expressed in megabytes (MB), what is the correct value according to the given options?

A. 2 MB	B. 20 MB
C. 200 MB	D. 2000 MB

Correct answer: D. 2000 MB

Reasoning: The model-paper options use the decimal convention 1 GB = 1000 MB, so 2 GB = 2000 MB. (Binary GiB/MiB conventions use 1024, but that is not the convention intended by these choices.)

Exam focus: Use the conversion intended by the given options.

MCQ 17

Which factor determines the validity of a scientific theory?

A. Length of time	B. Experimental evidence
C. Popularity	D. Number of equations

Correct answer: B. Experimental evidence

Reasoning: A scientific theory is judged by how well it agrees with reproducible observations and experiments, not by popularity or age.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 18

In a hydraulic press, the pressure applied on the small piston is:

A. Decreased	B. Transmitted equally
C. Increased	D. Vanished

Correct answer: B. Transmitted equally

Reasoning: Pascal's law gives $P1 = P2$ for the enclosed fluid. The force can increase at the larger piston because $F = PA$ and its area is larger.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 19

The energy stored in a compressed spring is:

A. Kinetic energy	B. Elastic potential energy
C. Chemical energy	D. Thermal energy

Correct answer: B. Elastic potential energy

Reasoning: A stretched or compressed spring stores elastic potential energy: $E = 1/2\ kx^2$.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 20

The slope of a distance-time graph gives:

A. Acceleration	B. Force
C. Speed	D. Work

Correct answer: C. Speed

Reasoning: Slope = Δdistance/Δtime, which is speed. A constant slope therefore represents uniform speed.

Exam focus: Graph slope meaning depends on the graph: distance-time -> speed; speed-time -> acceleration.

Momentum, Efficiency and Force

MCQ 21

A shopping cart is pushed with a force of 2 N for 5 s. What impulse is given to the cart?

A. 2 N s	B. 5 N s
C. 10 N s	D. 20 N s

Correct answer: C. 10 N s

Reasoning: Impulse J = FΔt = 2 x 5 = 10 N s. Impulse is also equal to change in momentum.

Exam focus: Impulse = force x time = change in momentum.

MCQ 22

A factory machine has an efficiency of 30%. It means that:

A. 30% of input energy is lost as heat	B. 30% of input energy is converted into useful output
C. 70% of input energy is converted into useful output	D. 40% of input energy is converted into useful output

Correct answer: B. 30% of input energy is converted into useful output

Reasoning: Efficiency = useful output/input x 100%. Therefore 30% efficiency means 30% of the input energy becomes useful output.

Exam focus: Efficiency describes the useful fraction of input energy.

MCQ 23

When pressure is applied on a surface, the force acts:

A. Parallel to the surface	B. Perpendicular to the surface
C. At 45 degrees to the surface	D. In the same direction as gravity

Correct answer: B. Perpendicular to the surface

Reasoning: Pressure is normal force per unit area: P = F_perpendicular/A. Therefore the force due to pressure acts perpendicular to the surface.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 24

When a bus starts suddenly, passengers fall backward due to:

A. Friction	B. Weight
C. Inertia	D. Gravitation

Correct answer: C. Inertia

Reasoning: The feet move forward with the bus while the upper body tends to remain in its initial state of rest. Relative to the bus, the passenger appears to fall backward.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 25

An object weighs 50 N on a planet and has mass 20 kg. What is the gravitational field strength on that planet?

A. 0.4 N/kg	B. 9.8 N/kg
C. 2.5 N/kg	D. 25 N/kg

Correct answer: C. 2.5 N/kg

Reasoning: Using W = mg, g = W/m = 50/20 = 2.5 N/kg.

Exam focus: Gravitational field strength g = W/m.

MCQ 26

Like parallel forces act in:

A. Opposite directions	B. Same direction
C. At 90 degrees to each other	D. At 45 degrees to each other

Correct answer: B. Same direction

Reasoning: Like parallel forces act in the same direction. Unlike parallel forces act in opposite directions.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 27

The state of matter found in stars is called:

A. Solid	B. Plasma
C. Gas	D. Liquid

Correct answer: B. Plasma

Reasoning: The very high temperature in stars ionizes matter into positive ions and free electrons. This ionized state is plasma.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 28

How much work is needed to lift a bag that weighs 150 N to a height of 4 m?

A. 300 J	B. 450 J
C. 600 J	D. 750 J

Correct answer: C. 600 J

Reasoning: W = Fd = 150 x 4 = 600 J.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 29

Vector X points east and vector Y points north. In which direction does the resultant lie?

A. North-East	B. South-East
C. North	D. East

Correct answer: A. North-East

Reasoning: The resultant of an eastward and a northward vector lies between those directions, i.e., toward North-East.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 30

The branch of physics that deals with the study of light is:

A. Mechanics	B. Optics
C. Thermodynamics	D. Magnetism

Correct answer: B. Optics

Reasoning: Optics is the branch of physics concerned with light and its behavior, including reflection and refraction.

Exam focus: Understand why the other options do not match the definition or physical law.

Motion, Magnetism and Measurements

MCQ 31

In which situation are the distance traveled and magnitude of displacement equal?

A. A child moving on a swing	B. An object moving along a circular path
C. A car moving on a curved road	D. Linear motion in one direction

Correct answer: D. Linear motion in one direction

Reasoning: Distance equals the magnitude of displacement only when the object moves along a straight line without reversing direction.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 32

When an external magnetic field is removed, a paramagnetic material usually:

A. Remains permanently magnetized	B. Loses most of its induced magnetism
C. Becomes strongly magnetic	D. Repels magnetic fields strongly

Correct answer: B. Loses most of its induced magnetism

Reasoning: Paramagnetic materials are only weakly magnetized in an external field. When the field is removed, most of the induced alignment is lost.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 33

In the given distance-time graph, portion A is a straight rising line. Which motion is displayed by the cyclist in portion A?

A. At rest	B. Uniform speed
C. Non-uniform speed	D. Variable speed

Correct answer: B. Uniform speed

Reasoning: A straight rising line on a distance-time graph has constant slope. Since slope of a distance-time graph is speed, portion A represents uniform speed.

Exam focus: Straight rising distance-time line means uniform speed, not acceleration.

MCQ 34

In scientific notation, 0.005 is written as:

A. 5 x 10^-3	B. 5 x 10^3
C. 5 x 10^-2	D. 5 x 10^-4

Correct answer: A. 5 x 10^-3

Reasoning: Moving the decimal three places to the right gives 5, so the power of ten is -3: 0.005 = 5 x 10^-3.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 35

A raindrop reaches a constant speed before hitting the ground. Why has it reached terminal velocity?

A. Its acceleration becomes zero	B. Its weight becomes zero

C. Air resistance disappears	D. Gravity no longer acts on it

Correct answer: A. Its acceleration becomes zero

Reasoning: At terminal velocity, air resistance balances weight, so net force is zero. Hence acceleration is zero and the drop continues at constant speed.

Exam focus: Terminal velocity occurs when net force and acceleration become zero.

MCQ 36

A spanner of length 30 cm uses a force of 150 N to open a tight nut. The torque is:

A. 10 N m	B. 25 N m
C. 35 N m	D. 45 N m

Correct answer: D. 45 N m

Reasoning: Assuming the force is perpendicular: r = 0.30 m and $\tau = Fr = 150 \times 0.30 = 45$ N m.

Exam focus: Convert centimeters to meters before calculating torque.

MCQ 37

What is a reasonable estimate for the mass of an apple?

A. 2 kg	B. 200 g
C. 2 mg	D. 20 kg

Correct answer: B. 200 g

Reasoning: An ordinary apple has a mass of a few hundred grams, making 200 g a reasonable estimate.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 38

A relationship between physical quantities remains unchanged under the same conditions after many experiments. Scientists accept it as a:

A. Theory	B. Assumption
C. Hypothesis	D. Law

Correct answer: D. Law

Reasoning: A scientific law describes a repeatedly observed relationship that remains consistent under stated conditions.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 39

Which physical property can be used to measure temperature?

A. Mass of a liquid	B. Volume of a liquid

C. Mass of a substance	D. Weight of a substance

Correct answer: B. Volume of a liquid

Reasoning: Liquid-in-glass thermometers rely on thermal expansion: the volume of the thermometric liquid changes with temperature.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 40

A perpetual motion machine is impossible because of:

A. Friction	B. Conservation of energy
C. Gravity	D. Air resistance

Correct answer: B. Conservation of energy

Reasoning: A machine cannot continuously produce energy without an equivalent input. That would violate conservation of energy.

Exam focus: Understand why the other options do not match the definition or physical law.

Scenario-Based MCQs: Toy Car

MCQ 41

Scenario: A remote-controlled car starts moving and gradually speeds up in a straight line. Which option best describes its motion?

A. Constant velocity	B. Constant acceleration
C. Decreasing velocity	D. Decreasing momentum

Correct answer: B. Constant acceleration (best matching option)

Reasoning: The scenario says the car speeds up, so it is accelerating. Among the given choices, constant acceleration is the best match, although the text itself does not explicitly state that the acceleration is perfectly constant.

Exam focus: Choose the best answer from the provided options and use the scenario wording carefully.

MCQ 42

How would the toy car's speed-time graph look, using the model-paper interpretation in Q41?

A. Straight line with negative slope	B. Straight line with positive slope
C. Curved graph	D. Horizontal line

Correct answer: B. Straight line with positive slope

Reasoning: For constant positive acceleration, $v = u + at$, so speed increases linearly with time. The speed-time graph is a straight line with positive slope.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 43

The toy car has mass 0.5 kg and acceleration 0.4 m/s^2. What force is applied?

A. 0.2 N	B. 0.4 N
C. 0.8 N	D. 2 N

Correct answer: A. 0.2 N

Reasoning: F = ma = 0.5 x 0.4 = 0.20 N.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 44

If the car mass is increased while the same force is applied, what happens to acceleration?

A. Increases	B. Decreases
C. Remains same	D. Becomes zero

Correct answer: B. Decreases

Reasoning: From a = F/m, for a constant force acceleration is inversely proportional to mass. Increasing mass decreases acceleration.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 45

What key idea should a worker keep in mind while operating a scrapyard crane with an electromagnet?

A. Electromagnet works without electricity	B. Magnet can be controlled ON/OFF
C. Electromagnet only attracts plastic materials	D. Magnet becomes weaker as iron is attached to it

Correct answer: B. Magnet can be controlled ON/OFF

Reasoning: An electromagnet depends on electric current, so its magnetic effect can be switched on for lifting and off for releasing.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 46

Why does the crane attract iron scraps and not plastic or wood?

A. Plastic and wood are lighter than iron	B. Plastic and wood are electrically charged
C. Iron is a magnetic material	D. Plastic and wood block magnetic field

Correct answer: C. Iron is a magnetic material

Reasoning: Iron is ferromagnetic and is strongly attracted by the electromagnet. Ordinary wood and plastic are not ferromagnetic.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 47

What happens if the electromagnet is replaced by a permanent bar magnet?

A. It will only attract iron pieces	B. It works as normal
C. It will attract only plastic materials	D. It will repel all types of materials

Correct answer: A. It will only attract iron pieces

Reasoning: Among the given options, A is correct. A permanent magnet will still attract iron, but it cannot be switched off easily, so practical lifting and releasing would be difficult.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 48

Why is an electromagnet preferred over a permanent magnet in a scrapyard crane?

A. It is cheaper than a permanent magnet	B. It allows control over lifting and releasing scrap
C. It attracts all metals permanently	D. It does not use electricity

Correct answer: B. It allows control over lifting and releasing scrap

Reasoning: Current can be switched on to lift magnetic scrap and switched off to release it.

Exam focus: Understand why the other options do not match the definition or physical law.

Scenario-Based MCQs: Principle of Moments

MCQ 49

The beam, pivot, sandbag and carpenter balancing arrangement is an example of:

A. Circular motion	B. Principle of moments
C. Principle of inertia	D. Principle of pulley

Correct answer: B. Principle of moments

Reasoning: For rotational equilibrium, clockwise moment equals anticlockwise moment about the pivot.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 50

Which factor does NOT affect the moment of a force?

A. Shape of the object	B. Magnitude of force
C. Distance from pivot	D. Direction of force

Correct answer: A. Shape of the object

Reasoning: Moment depends on force, perpendicular distance from the pivot, and force direction ($\tau = rF \sin\theta$). The shape itself is not a direct factor.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 51

A 5 kg sandbag is 1.2 m to the left of the pivot. The carpenter weighs 300 N. How far to the right must he sit to balance? (g = 10 m/s^2)

A. 0.8 m	B. 1.2 m
C. 2 m	D. 0.2 m

Correct answer: D. 0.2 m

Reasoning: Sandbag weight = 5 x 10 = 50 N. Balance condition: 50 x 1.2 = 300 x x. Thus x = 60/300 = 0.2 m.

Exam focus: The diagram supplies the carpenter weight as 300 N.

MCQ 52

The beam is 4 m long and the pivot is shifted so that it is 1 m from the sandbag at the left end. What is the carpenter moment arm at the right end?

A. 1 m	B. 2 m
C. 3 m	D. 4 m

Correct answer: C. 3 m

Reasoning: Moment arm = distance from pivot to carpenter = 4 - 1 = 3 m.

Exam focus: Understand why the other options do not match the definition or physical law.

Scenario-Based MCQs: Satellite and Rocket

MCQ 53

Which expression is used to determine the orbital velocity of a satellite around Earth?

A. $2\pi r/t$	B. $2\pi r/T$
C. $2\pi T/R$	D. $2\pi t/r$

Correct answer: B. $2\pi r/T$

Reasoning: Orbital speed = distance traveled in one orbit / orbital period = circumference / period = $2\pi r/T$.

Exam focus: One orbit distance is the circumference $2\pi r$.

MCQ 54

A satellite has orbital period 96.5 min and orbital radius 6.97 x 10^6 m. What is its orbital speed?

A. 7.56 km/s	B. 7.56 m/s
C. 756 km/s	D. 7.56 cm/s

Correct answer: A. 7.56 km/s

Reasoning: T = 96.5 x 60 = 5790 s. v = $2\pi r/T = 2\pi(6.97 \times 10^6)/5790 \approx 7.56 \times 10^3$ m/s = 7.56 km/s.

Exam focus: Convert minutes into seconds before using SI units.

MCQ 55

A rocket expels hot gases downward. Which force directly pushes the rocket upward?

A. Friction between rocket and air	B. Force of engine downward
C. Exhaust gases pushing against the ground	D. Reaction force exerted by exhaust gases on the rocket

Correct answer: D. Reaction force exerted by exhaust gases on the rocket

Reasoning: The rocket pushes exhaust gases backward/downward; the gases exert an equal and opposite force on the rocket. This is Newton's third law.

Exam focus: Rocket thrust is a Newton's third-law interaction with exhaust gases, not with the ground.

MCQ 56

Which principle explains the working of a rocket in space?

A. Pascal's law	B. Buoyant force
C. Newton's third law of motion	D. Thermal expansion of gases

Correct answer: C. Newton's third law of motion

Reasoning: A rocket works by expelling gases one way and receiving an equal and opposite reaction. It does not need air or the ground to push against.

Exam focus: Understand why the other options do not match the definition or physical law.

Scenario-Based MCQs: Pressure and Snowshoes

MCQ 57

Why do flat snowshoes help a person walk on snow without sinking deeply?

A. They increase weight on snow	B. They decrease weight on snow
C. They increase contact area, decreasing pressure	D. They decrease contact area, decreasing pressure

Correct answer: C. They increase contact area, decreasing pressure

Reasoning: P = F/A. For the same weight, increasing contact area decreases pressure on the snow.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 58

For a constant force, what is the relation between pressure and area of contact?

A. Direct	B. Exponential
C. Proportional	D. Inverse

Correct answer: D. Inverse

Reasoning: P = F/A. With F constant, P is inversely proportional to A.

Exam focus: Understand why the other options do not match the definition or physical law.

MCQ 59

Amir weighs 600 N and the contact area of his sports shoes is 0.02 m^2. What pressure does he exert on snow?

A. 12 kPa	B. 30 kPa
C. 600 Pa	D. 30 Pa

Correct answer: B. 30 kPa

Reasoning: P = F/A = 600/0.02 = 30,000 Pa = 30 kPa.

Exam focus: 1 kPa = 1000 Pa.

MCQ 60

Amir's father has the same weight (600 N) but snowshoe area 0.20 m^2. How many times smaller is his pressure?

A. 20 times	B. 2 times
C. 5 times	D. 10 times

Correct answer: D. 10 times

Reasoning: Father: P = 600/0.20 = 3000 Pa. Amir: 30,000 Pa. Ratio = 30,000/3000 = 10, so the father exerts 10 times less pressure.

Exam focus: For the same force, pressure ratio is the inverse of area ratio.

Section B - Complete Short-Question Solutions

The model paper asks six brief answers of 3 marks each. For preparation, every main question and every OR alternative is solved below.

Q1. Differentiate between base quantities and derived quantities with one example of each.

Definition: Base quantities are fundamental physical quantities that are independent of other physical quantities. Derived quantities are obtained by mathematical combinations of base quantities.

Examples: Base quantity example: mass, SI unit kg. Derived quantity example: force, F = ma, SI unit N = kg m/s^2.

Full-marks conclusion: Base quantities are fundamental and independent, whereas derived quantities are expressed in terms of base quantities.

Full-marks presentation: Use the law/formula, show the logical steps, and end with a clear final statement or boxed numerical answer.

OR: Differentiate between mass and weight (any three points).

Mass: Amount of matter in a body; scalar; SI unit kg; approximately constant from place to place; measured with a balance.

Weight: Gravitational force on a body; vector; SI unit N; changes with gravitational field strength; measured with a spring balance.

Relation: W = mg. For full marks, write at least three paired differences clearly.

Q2. A car starts from rest and reaches 20 m/s in 5 s. Calculate its average velocity and total distance covered.

Given: u = 0 m/s, v = 20 m/s, t = 5 s. Assume uniform acceleration.

Average velocity: v_avg = (u + v)/2 = (0 + 20)/2 = 10 m/s.

Distance: s = v_avg t = 10 x 5 = 50 m.

Answer: Average velocity = 10 m/s; total distance = 50 m.

Full-marks presentation: Use the law/formula, show the logical steps, and end with a clear final statement or boxed numerical answer.

OR: Prove that pressure of a liquid of density rho at depth h is P = rho g h.

Step 1: Consider a liquid column of height h and cross-sectional area A. Its volume V = Ah.

Step 2: Mass m = rho V = rho Ah.

Step 3: Weight (force) F = mg = rho Ahg.

Step 4: Pressure P = F/A = rho Ahg/A = rho gh.

Result: P = rho gh. Therefore liquid pressure increases with density and depth.

Q3. A bullet is fired from a gun. It penetrates into sand and then stops. Where is its kinetic energy used?

Initial energy: The moving bullet has kinetic energy KE = $1/2\ mv^2$.

During penetration: The bullet does work against resistive/frictional forces of the sand.

Energy transformations: Its kinetic energy is mainly converted into internal (thermal) energy of the bullet and sand, deformation/displacement of sand, and a small amount of sound and vibration.

Conservation: The energy is not destroyed; it changes form according to conservation of energy.

Full-marks presentation: Use the law/formula, show the logical steps, and end with a clear final statement or boxed numerical answer.

OR: Why is the spring constant of a steel spring greater than that of a rubber spring with the same number of turns and cross-sectional area?

Hooke's law: $F = kx$, so $k = F/x$.

Material property: Steel is much stiffer and has a much larger elastic modulus than rubber.

Reasoning: For the same applied force, steel extends less than rubber. Since $k = F/x$, smaller extension means larger spring constant.

Result: $k_{steel} > k_{rubber}$ because of the different elastic properties of the materials.

Q4. Can a body have zero velocity but non-zero acceleration? Explain with a physical example.

Answer: Yes.

Example: Throw a ball vertically upward. At the highest point, its instantaneous velocity is $v = 0$.

Acceleration: Gravity is still acting, so acceleration is $a = -g \approx -9.8\ m/s^2$ (taking upward as positive).

Conclusion: Therefore a body can have zero velocity at an instant while having non-zero acceleration.

Full-marks presentation: Use the law/formula, show the logical steps, and end with a clear final statement or boxed numerical answer.

OR: A solenoid has length 50 cm and 100 turns. Calculate its magnetic field if 1.2 A current passes through it.

Given: L = 50 cm = 0.50 m, N = 100 turns, I = 1.2 A.

Turns per metre: $n = N/L = 100/0.50 = 200\ m^{-1}$.

Formula: For an air-core solenoid, $B = \mu_0 n I$, where $\mu_0 = 4\pi \times 10^{-7}$ T m/A.

Calculation: $B = (4\pi \times 10^{-7})(200)(1.2) \approx 3.02 \times 10^{-4}$ T.

Answer: $B \approx 3.0 \times 10^{-4}$ T = 0.30 mT. If the term magnetic field strength is used strictly for H, then $H = nI$ = 240 A/m.

Q5. Explain motion of a rocket using the law of conservation of momentum.

Initial state: Before firing, the total momentum of the rocket-fuel system may be taken as zero if it is initially at rest.

Gas ejection: Burning fuel produces hot gases that are expelled backward at high speed, giving the gases backward momentum.

Conservation: To keep total momentum conserved, the rocket gains equal forward momentum: $p_{rocket} + p_{gas} = \text{constant}$.

Conclusion: Thus the rocket accelerates forward/upward. It does not need air to push against, so it can also work in space.

Full-marks presentation: Use the law/formula, show the logical steps, and end with a clear final statement or boxed numerical answer.

OR: How can we increase sensitivity and range of a liquid-in-glass thermometer?

Sensitivity: Sensitivity is the change in liquid-column length per degree change in temperature.

Increase sensitivity: Use a narrower capillary tube, a larger bulb, and a thermometric liquid with a larger coefficient of expansion.

Range: Range is the interval between the minimum and maximum measurable temperatures.

Increase range: Use a longer stem/capillary, choose a liquid with a low freezing point and high boiling point, and use suitable bore/bulb dimensions. Note that designs that increase sensitivity may reduce range, so a compromise is needed.

Q6. What happens to a disc rotating freely in space when no external moment acts on it? Give a reason.

Result: The disc continues rotating with constant angular momentum.

Reason: External torque $\tau_{ext} = 0$. Since $\tau = dL/dt$, $dL/dt = 0$, so angular momentum L remains constant.

If shape is unchanged: $L = I\omega$. If moment of inertia I stays constant, angular velocity ω also stays constant.

Conclusion: Therefore the disc keeps rotating at constant angular velocity in the absence of external torque.

Full-marks presentation: Use the law/formula, show the logical steps, and end with a clear final statement or boxed numerical answer.

OR: Two identical balls roll on a smooth floor. Ball A has twice the momentum of ball B. The same force stops both. Which ball takes more time to stop?

Given: $p_A = 2p_B$ and the stopping force F is the same.

Impulse-momentum theorem: $F\Delta t = \Delta p$. To stop a ball, the magnitude of the required change in momentum equals its initial momentum.

Comparison: $t_A = p_A/F = 2p_B/F = 2t_B$.

Answer: Ball A takes twice as much time to come to rest.

Section C - Complete Long-Question Solutions

Each long question carries 6 marks. The answers below are arranged in clear sub-points so students can reproduce them systematically in the examination.

Q7. Suggest any three ways to magnetize and demagnetize a material.

Magnetization - Stroking method: Stroke an iron or steel bar repeatedly in the same direction with one pole of a strong magnet. The magnetic domains become more aligned.

Magnetization - Electrical/solenoid method: Place the magnetic material inside a solenoid and pass direct current through it. The solenoid field aligns domains in the material.

Magnetization - Induction method: Bring the material into a strong magnetic field. Magnetic induction produces domain alignment and temporary or permanent magnetization depending on the material.

Demagnetization - Heating: Strong heating increases thermal agitation and disturbs domain alignment, weakening or destroying magnetism.

Demagnetization - Mechanical shock: Repeated hammering or dropping can disturb domain alignment and reduce magnetization.

Demagnetization - Alternating field: Place the magnet in a solenoid carrying AC and gradually reduce the field or slowly withdraw the magnet. The reversing field randomizes domains.

Core idea: Magnetization mainly involves domain alignment; demagnetization involves loss of ordered domain alignment.

OR: Describe three states of equilibrium. Give one clear daily-life example of each.

Stable equilibrium: After a small displacement the body tends to return to its original position. Its center of gravity rises on displacement and potential energy increases. Example: a ball at the bottom of a bowl.

Unstable equilibrium: After a small displacement the body moves farther away from its original position. Its center of gravity falls and potential energy decreases. Example: a ball balanced on top of a hill/dome.

Neutral equilibrium: After displacement the body remains in its new position. The center of gravity stays at the same height and potential energy is unchanged. Example: a ball on a horizontal floor.

Equilibrium condition: For complete mechanical equilibrium: resultant force = 0 and resultant torque = 0.

Q8. Define acceleration. What does it tell us about motion? What does a steeper slope of a speed-time graph indicate?

Definition: Acceleration is the rate of change of velocity with time: $a = \Delta v/\Delta t = (v - u)/t$. SI unit: m/s^2.

Meaning: Acceleration tells how quickly velocity changes. Velocity can change because of a change in speed, direction, or both.

Graph relation: For a speed-time graph, slope = Δspeed/Δtime, so the slope represents acceleration (for one-dimensional speed increase).

Steeper slope: A steeper upward slope means a greater acceleration because speed is changing more rapidly with time.

Exam conclusion: Steeper speed-time line => larger acceleration, provided the graph is being compared over the same axes/scales.

OR: Give two real-life examples of energy conversion, explain each mechanism, and explain why some energy is transferred to surroundings as heat.

Example 1 - Electric fan: Electrical energy is converted mainly into kinetic/mechanical energy of the rotating blades by the electric motor. Some energy becomes heat in the coils because of electrical resistance, and some becomes sound/vibration.

Example 2 - Falling object: Gravitational potential energy mgh is converted into kinetic energy as the object falls. With air resistance and during impact, part of the energy becomes thermal energy, sound and deformation.

Why heat is produced: Friction, air resistance, electrical resistance and deformation transfer part of the energy into random microscopic motion of particles, which appears as thermal energy.

Conservation: Energy is not destroyed. Total energy before and after remains the same, but some becomes less useful because it is dispersed to the surroundings.

Final Revision - Formula Sheet

Quantity / Principle	Formula
Speed	v = distance/time
Velocity	v = displacement/time
Acceleration	$a = (v - u)/t$
Uniform acceleration average velocity	$v_{avg} = (u + v)/2$
Distance using average velocity	$s = v_{avg} t$
Newton's second law	$F = ma$
Momentum	$p = mv$
Impulse	$J = F\Delta t = \Delta p$
Work	$W = Fd$ (for force parallel to displacement)
Gravitational potential energy	$PE = mgh$
Kinetic energy	$KE = 1/2\ mv^2$
Pressure	$P = F/A$
Liquid pressure	$P = \rho gh$
Weight	$W = mg$
Gravitational field strength	$g = W/m$
Torque / moment	$\tau = F d_{perpendicular}$
Principle of moments	Sum clockwise moments = sum anticlockwise moments
Spring relation	$F = kx$
Elastic potential energy	$E = 1/2\ kx^2$
Efficiency	η = useful output/input x 100%
Orbital speed	$v = 2\pi r/T$
Solenoid field	$B = \mu_0 nI$, where $n = N/L$
Angular momentum	$L = I\omega$

Quantity / Principle	Formula
Torque and angular momentum	$\tau = dL/dt$

Last-Minute Exam Strategy

- MCQs: identify the governing definition or formula before looking at distractors.
- Numericals: convert all quantities to SI units before substitution.
- Always write units with final numerical answers.
- For graph questions, identify what the slope represents on that specific graph.
- For OR questions, prepare both during revision even though only one is attempted in the exam.
- For 6-mark answers, use headings or numbered points so the examiner can see each required part clearly.

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